

Cardiac Valve Pathology

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All the images and information included in this presentation are from
Robbins and Cotran, Pathologic Basis of Disease, 10th edition

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Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

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At the conclusion of the session, you will be able to:

- Describe and explain the most important features defining cardiac valvular disease
- Compare and contrast the causes and consequences of **aortic valve stenosis**
- Define **mitral valve prolapse and mitral annular calcification**
- Demonstrate an understanding of the etiopathogenesis of **acute rheumatic fever and rheumatic heart disease**
- Summarize the pathology of **acute and subacute infective endocarditis**, as well as their associated macroscopic, microscopic, and clinical consequences
- Differentiate between **nonbacterial endocarditis and Libman-Sacks endocarditis**

Valvular disease

Valvular disease may result in stenosis, or both.

- **Stenosis:** it is caused by a **primary cuspal abnormality** determined by a chronic process: calcification or valve scarring.
- **Insufficiency:** is caused by an **intrinsic disease of the valve cusps** (e.g., endocarditis) or **disruption of the supporting structures** (tendinous cords, papillary muscles).

It can appear abruptly, as with chordal rupture, or insidiously as a consequence of leaflet scarring and retraction

Valvular disease can involve only one valve (the mitral valve being the most common target), or more than one valve.

Turbulent flow through diseased valves → murmurs

Severe stenosis → thrills (may be palpated)

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The Outcome: depends on:

- the valve involved
- the degree of impairment
- the tempo of its development
- the compensatory mechanisms.

Rapid destruction of an aortic valve cusp by infection can cause massive regurgitation and the abrupt onset of cardiac failure.

Rheumatic mitral stenosis usually progresses over years, and its clinical effects are well tolerated until late in the course.

Etiology

Valvular abnormalities can be congenital or acquired

The most common congenital valvulopathy is a bicuspid aortic valve, containing only two functional cusps instead of the normal three

Acquired valvulopathies: stenoses of the aortic and mitral valves account for approximately two thirds of all valve disease.

Mitral Valve Disease	Aortic Valve Disease
Mitral Stenosis	Aortic Stenosis
Postinflammatory scarring (rheumatic heart disease)	Postinflammatory scarring (rheumatic heart disease)
	Senile calcific aortic stenosis
	Calcification of congenitally deformed valve
Mitral Regurgitation	Aortic Regurgitation
Abnormalities of leaflets and commissures	Intrinsic valvular disease
Postinflammatory scarring	Postinflammatory scarring (rheumatic heart disease)
Infective endocarditis	Infective endocarditis
Mitral valve prolapse	Aortic disease
“Fen-phen”—induced valvular fibrosis	Degenerative aortic dilation
Abnormalities of tensor apparatus	Syphilitic aortitis
Rupture of papillary muscle	Ankylosing spondylitis
Papillary muscle dysfunction (fibrosis)	Rheumatoid arthritis
Rupture of chordae tendineae	Marfan syndrome
Abnormalities of left ventricular cavity and/or annulus	
Left ventricular enlargement (myocarditis, dilated cardiomyopathy)	
Calcification of mitral ring	

Fen-phen, Fenfluramine-phentermine.

Calcific Aortic Stenosis

Most cases

- asymptomatic
- calcifications on a CXR

In some patients, it can be severe enough to require surgical intervention.

The incidence parallels longevity.

In normal valves: when patients reach their 70s and 80s;

In bicuspid valves: when patients reach their 40s and 50s

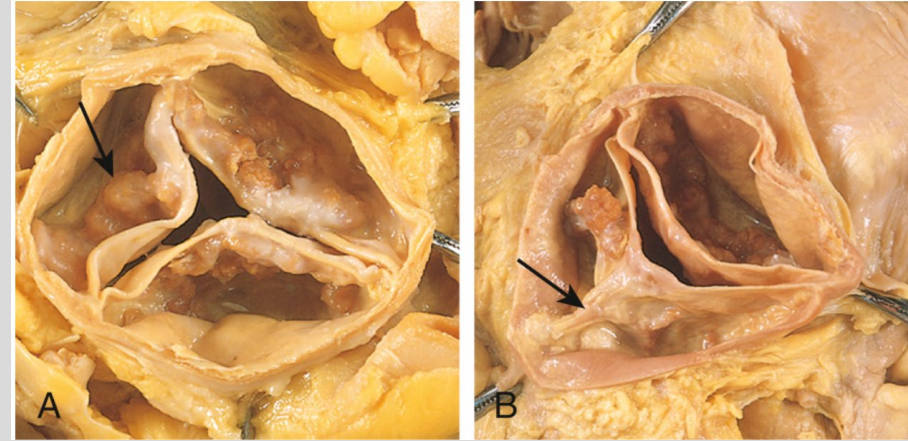
Pathogenesis

Simple progressive age-associated “wear and tear”

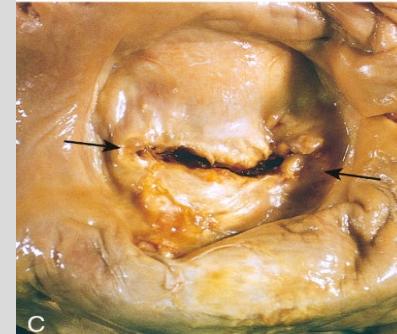
Cuspal fibrosis and calcification

Equivalent of atherosclerosis

It is associated with hyperlipidemia, hypertension, inflammation, and other factors implicated in atherosclerosis

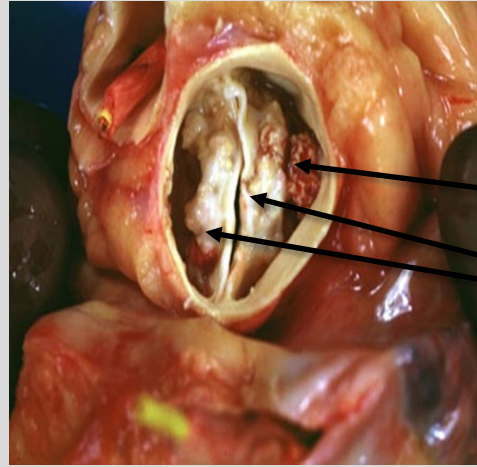
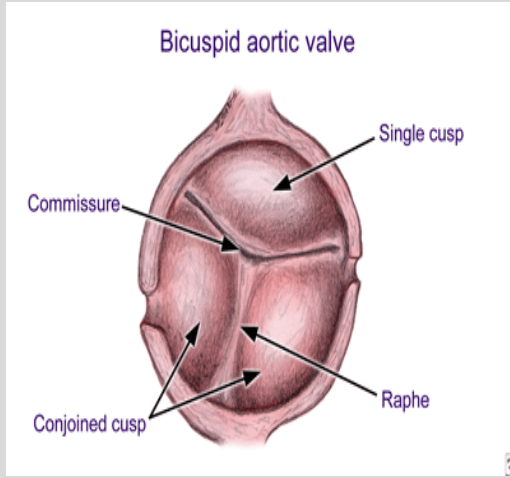


- A) Calcific aortic stenosis of a previously normal. Nodular masses of calcium are heaped up within the sinuses of Valsalva (arrow) .
- B) Calcific aortic stenosis occurring on a congenitally bicuspid valve. One cusp has a partial fusion at its center, called a raphe (arrow)



- C) Fused commissures in rheumatic aortic valve stenosis

Congenital Bicuspid Aortic Valve



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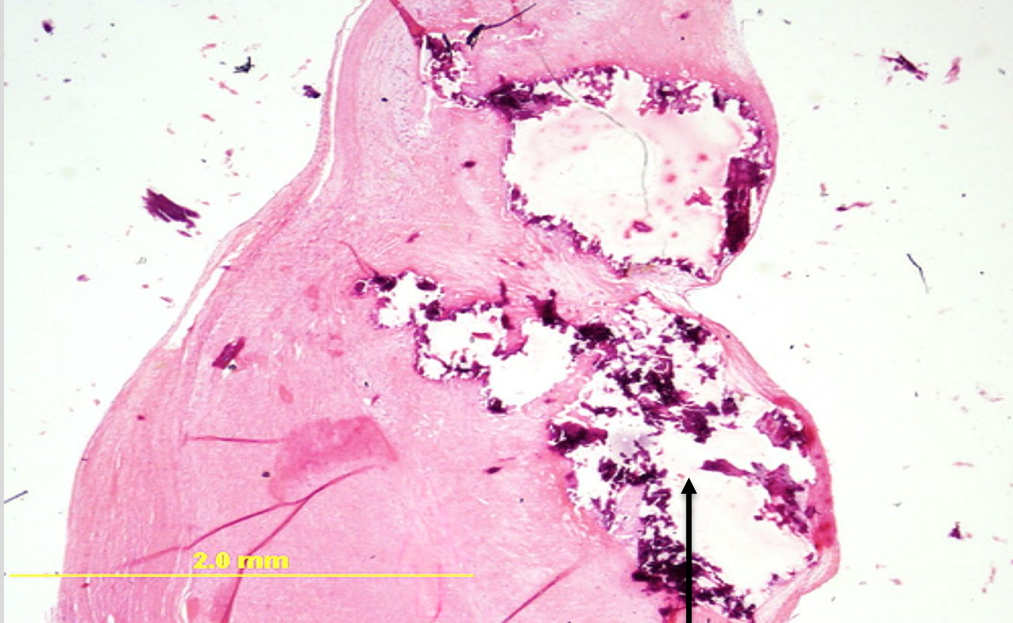
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thrombus

calcifications

- In ~1% of the population;
- ~50% of cases of aortic stenosis in adults
- Familial clustering → associated with mutation in NOTCH1
- Morphology: two functional cusps, usually unequal in size
 - Larger valve → has midline raphe (from incomplete commissural separation during development)
 - Raphe = major site where calcific deposits occur

Calcific Aortic Stenosis Histology



nodular calcific deposits

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The calcified masses generally start at the base.

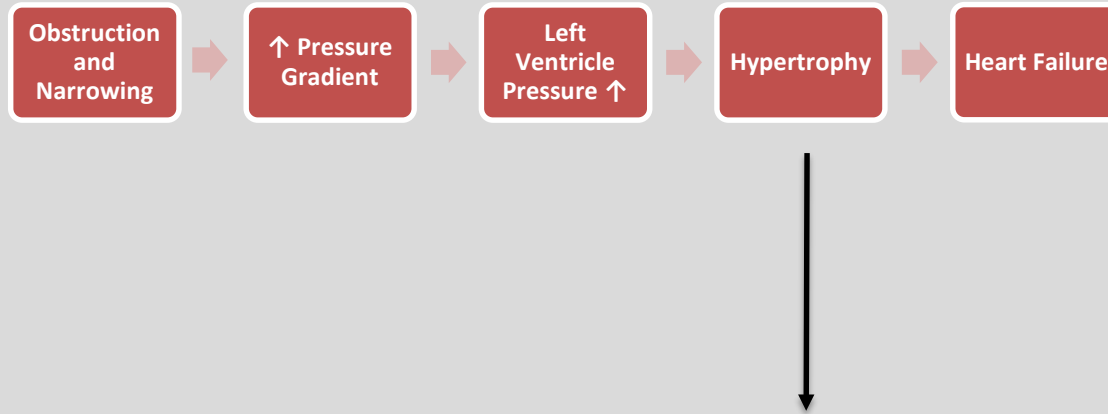
Calcium deposition leads to a reduction of the valve functional area which determines outflow obstruction and pressure overload.



Calcific Aortic Stenosis Physiopathology

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Ischemia, Angina, Syncope
Exhaustion of compensatory
cardiac function

Mitral Valve Prolapse (MVP)

One or both mitral leaflets are “floppy” and prolapse—they balloon back into the left atrium during systole.

Associated conditions

Marfan syndrome
Ehlers-Danlos syndrome
polycystic kidney disease
fragile X syndrome

Etiology

Myxomatous degeneration
-Primary spontaneous
-Secondary to a connective tissue disorder
Rheumatic fever
Myocardial infarction

Pathogenesis

myxomatous degeneration of the mitral valve

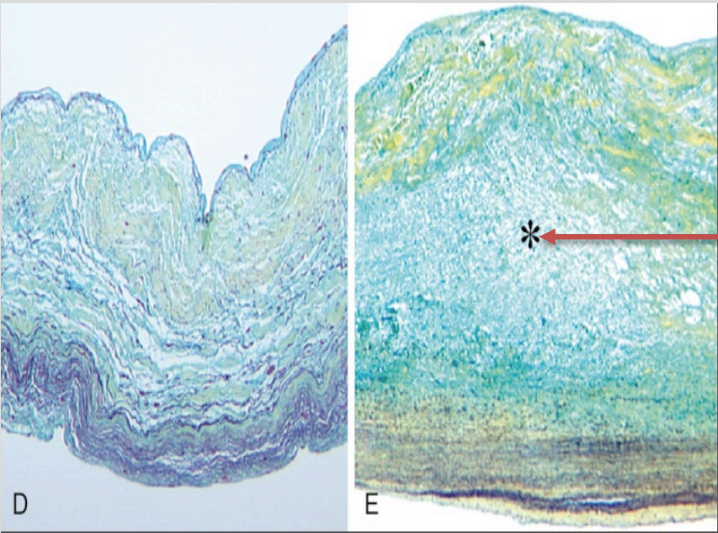
- valvular leaflets become abnormal secondary to
 - excessive degradation of collagen and elastin by proteolytic enzymes
 - accumulation of proteoglycans
- the chordae tendinae become weakened and elongated
- annular dilation of the mitral valve

connective tissue disorders

- deficiency in collagen and elastin cause leaflets to be mostly thin and translucent

Mitral Valve Prolapse Morphology

- prominent hooding with prolapse of the posterior mitral leaflet (arrow) into the left atrium;
- the atrium also is dilated suggesting long-standing valvular insufficiency and volume overload



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Normal heart valve (D) and myxomatous mitral valve (E).

E: collagen in the fibrosa layer is loose and disorganized, proteoglycan deposition in the central spongiosa layer is markedly expanded

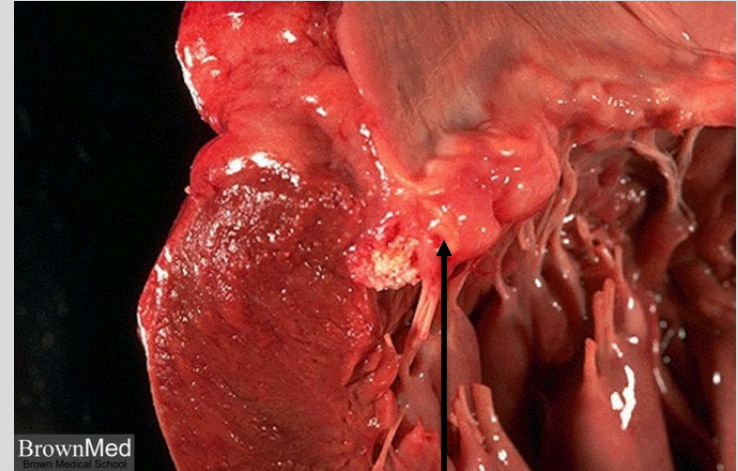
Mitral Annular Calcification (MAC)

- MAC is most commonly an incidental finding requiring no specific treatment other than assessment and modification of cardiovascular risk factors.
- See in women > 60 years old; especially in those with myxomatous mitral valve or elevated left ventricle pressure
- **Pathogenesis:** degenerative calcific deposits that develop in the fibrous annulus.
- MAC may be associated with atherosclerosis.

The most important intervention for most patients with MAC is evaluation and treatment of standard cardiovascular risk factors.

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Irregular, stony hard, occasionally ulcerated nodules at the base of the leaflets

Acute Rheumatic Fever (ARF) and Rheumatic Heart Disease (RHD)

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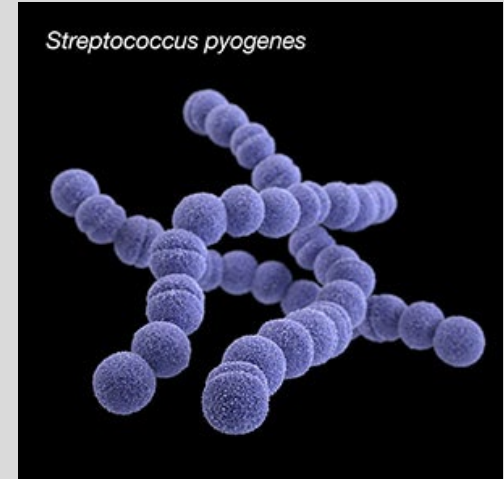
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Acute rheumatic fever RF is immunologically mediated, multisystem inflammatory disease that occurs after group A β -hemolytic streptococcal infections (pharyngitis, and occasionally infections at other sites, such as skin).

Rheumatic heart disease RHD is the cardiac manifestation of acute rheumatic fever.

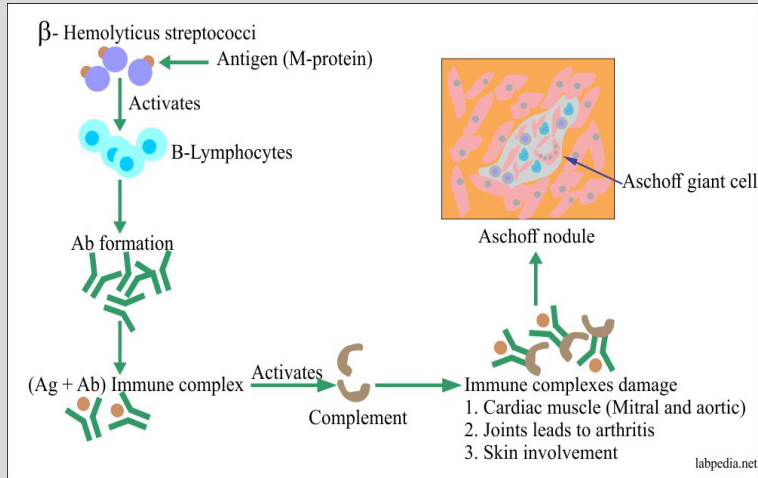
It is associated with inflammation of all parts of the heart, but valvular inflammation and scarring produce the most important clinical features.

Rheumatic heart disease is essentially the only cause of acquired mitral stenosis.



CDC

Pathogenesis



ARF is a host immune responses to group A streptococcal antigens that cross-react with host proteins.

1. Ab-Ag binding can activate Complement
2. Cytokine production by the stimulated T cells leads to macrophage activation

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RHD is an immunological reaction: streptococci are completely absent from the lesions.

A genetic susceptibility is likely to influence the development of the cross-reactive immune responses.

<https://labpedia.net/elementary-immunology/chapter-25-autoimmune-diseases-rheumatic-fever/>

ARF Clinical Features

3% of patients: typically appears 10 days to 6 weeks after a group A streptococcal infection.

It occurs most often in children between 5 and 15 years of age

Major Criteria for Dx

J - Joint Pain

♥ - Heart

N - Nodules

E - Erythema marginatum

S - Sydenham chorea

Acute carditis:

Pericardial friction rubs

Tachycardia

Arrhythmias.

Myocarditis

Mitral valve insufficiency

Heart failure.

1% of affected individuals die of fulminant RF involvement of the heart.

Arthritis typically begins with migratory polyarthritis (accompanied by fever) in which one large joint after another becomes painful and swollen for a period of days and then subsides spontaneously, leaving no residual disability

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Acute Rheumatic Fever

Subcutaneous nodules present on extensor surfaces



Semantic Scholar

Sydenham chorea : neurologic disorder with involuntary rapid movement (dance -like movements)



Sydenham's Chorea

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Erythema marginatum :
irregular skin rash areas



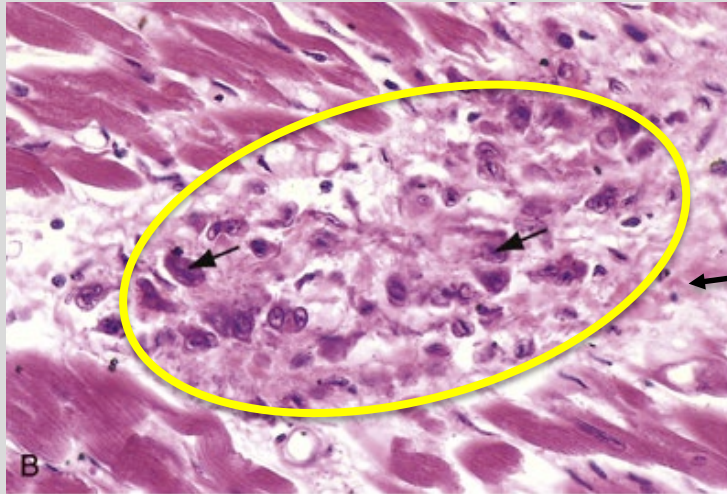
Dermatology Advisory

ARF Morphology

Myocarditis

Distinctive lesions in the heart—called **Aschoff bodies** —are composed of foci of T lymphocytes, occasional plasma cells, and plump activated macrophages called **Anitschkow cells**.

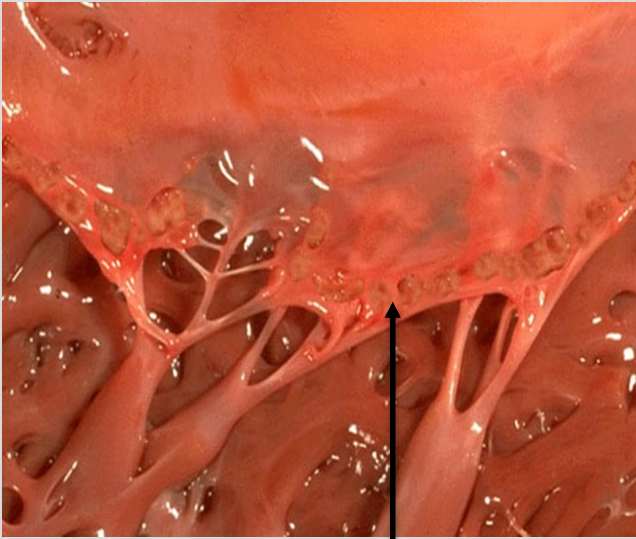
Diffuse inflammation and Aschoff bodies may be found in any of the three layers of the heart, resulting in pericarditis, myocarditis, or endocarditis (pancarditis).



Aschoff body in a patient with acute carditis.

The myocardium exhibits a circumscribed nodule of mixed mononuclear inflammatory cells with associated necrosis. Within the inflammation, large activated macrophages show prominent nucleoli, as well as chromatin condensed into long, wavy ribbons (caterpillar cells) represented by the arrows.

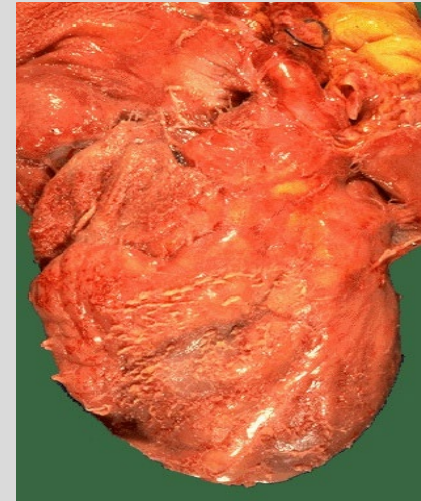
These are also known as Anitschkow cells.



Endocarditis – Irregular, warty projects/vegetations along the lines of closure

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Pericarditis – “bread and butter”
→ fibrinous pericardial exudate

RHD Clinical Features

It is diagnosed years or even decades after the initial episode of RF

Cardiac murmurs

Cardiac hypertrophy and dilation

Heart failure

Arrhythmias

Thromboembolic complications

Endocarditis

Variable prognosis

Surgical repair of diseased valves

Replacement of diseased valves

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Chronic Rheumatic Heart Disease Morphology

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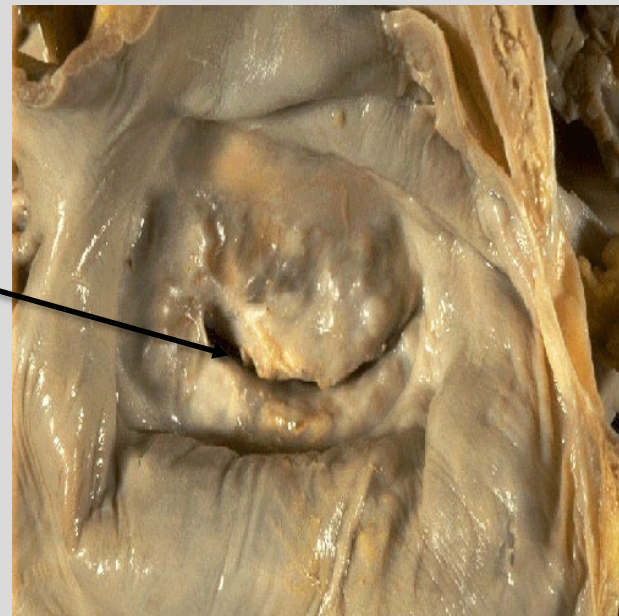
The mitral valve is virtually always involved in chronic RHD;
It is affected in isolation in roughly two-thirds of cases.
It is affected along with the aortic valve in another 25%.

Tricuspid valve and pulmonary valve are only rarely affected

In **rheumatic mitral stenosis**, calcification and fibrous bridging across the valvular commissures create “fish mouth” stenoses.

If severe, the left atrium progressively dilates and may harbor **mural thrombi** that can **embolize**.

Congestive changes in the lungs may induce pulmonary vascular and parenchymal changes; over time, these can lead to **right ventricular hypertrophy**.



Infective Endocarditis (IE)

=microbial infection of the heart valves or the mural endocardium that leads to the formation of vegetations composed of thrombotic debris and organisms, often associated with destruction of the underlying cardiac tissues.

.

Acute and subacute forms

Acute IE : **infection of a previously normal heart valve** by a highly virulent organism (e.g., *Staphylococcus aureus*) that rapidly produces destructive lesions.

- Difficult to cure with antibiotics alone and often require surgery
- Substantial morbidity and even mortality.

Subacute IE: **infection of a deformed valve** by organisms with lower virulence (e.g., viridans streptococci) that cause insidious infections;

- Overall less destruction.
- Protracted course of weeks to months
- Cures can often be achieved with antibiotics alone

Infective Endocarditis Risk Factors

- Rheumatic Heart Disease
- Mitral Valve Prolapse
- Degenerative Calcific Valvular Stenosis
- Prosthetic Valves
- Congenital Defects
- Immunocompromised patients (i.e., drug abuse disorder, neutropenia, immunodeficiency, malignancy, diabetes)

Infective Endocarditis Clinical Features

Acute endocarditis has a stormy onset with rapid onset of

- Fever
- Chills
- Weakness

Fever is the most consistent sign of IE; it can be slight or absent, in older adults.

Murmurs are present in the majority of patients, either from a new valvular defect or from a preexisting abnormality.

Infective Endocarditis Etiology

60% of cases : **Streptococcus viridans** (a commensal in the oral cavity)

Virulent S. aureus found on the skin can infect either **healthy or deformed valves**

-20% to 30% of cases overall

Other bacterial causes :-Enterococci

-HACEK group : -Haemophilus, Actinobacillus,
Cardiobacterium, Eikenella,
Kingella

-Commensals in the oral cavity.

Gram-negative bacilli and fungi : rarely

Prosthetic valve endocarditis occurring in the 1 to 2 months after surgical implantation is typically caused by skin flora , S. aureus and S. epidermidis ; prosthetic valve infections 1 year or more after surgery tend to be caused by streptococci.

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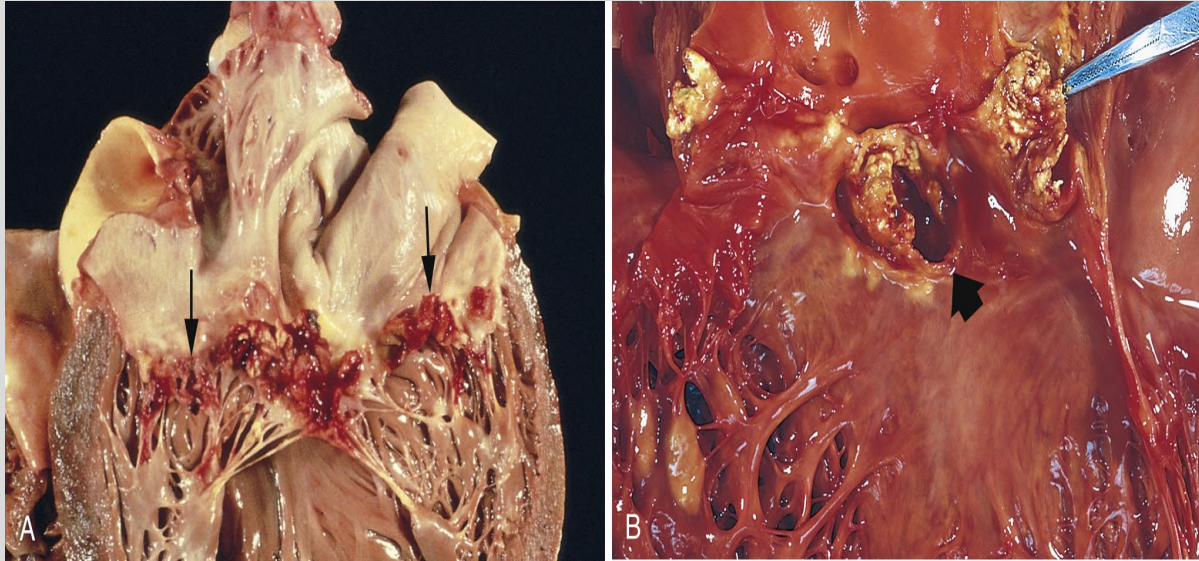


Infective Endocarditis Morphology: Vegetations

The aortic and mitral valves are the most common sites of infection

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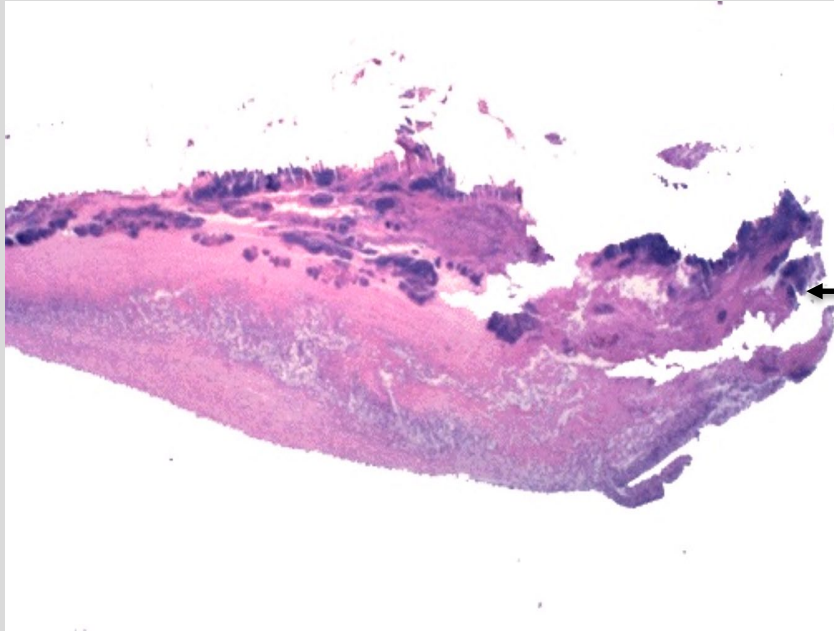
(A) Subacute IE of mitral valve : large, friable vegetations caused by *S. viridans* are indicated by arrows .

(B) Acute IE of congenitally bicuspid aortic valve caused by *Staphylococcus aureus* : extensive cuspal destruction and ring abscess (arrow) .

Infective Endocarditis Histology

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Vegetation:

fibrin

platelets

inflammatory cells

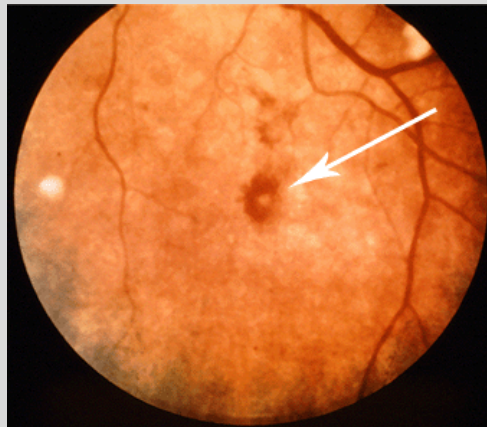
bacteria

Infective Endocarditis



Simple Cardiology

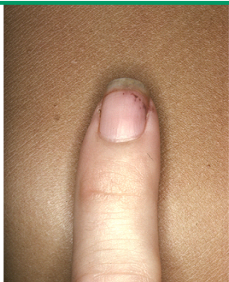
A. Janeway lesions



Merck Manual

B. Roth spots

Splinter hemorrhages in infective endocarditis



Splinter hemorrhages, linear reddish-brown lesions, are seen in the nail bed of this patient with bacterial endocarditis due to group B *Streptococcus*.

Courtesy of Gene Beyt, *The Skin and Infection: A Color Atlas and Text*, Sanders CV, Nesbitt LT Jr (Eds), Williams & Wilkins, Baltimore 1995.

<http://www.lww.com>

UpToDate®

C. Splinter hemorrhages



D. Osler nodes

<https://medpics.ucsd.edu/index.cfm?courcepage=image&course=clinlmg&mode=browse&lesson=118&img=2172>

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Nonbacterial Thrombotic Endocarditis

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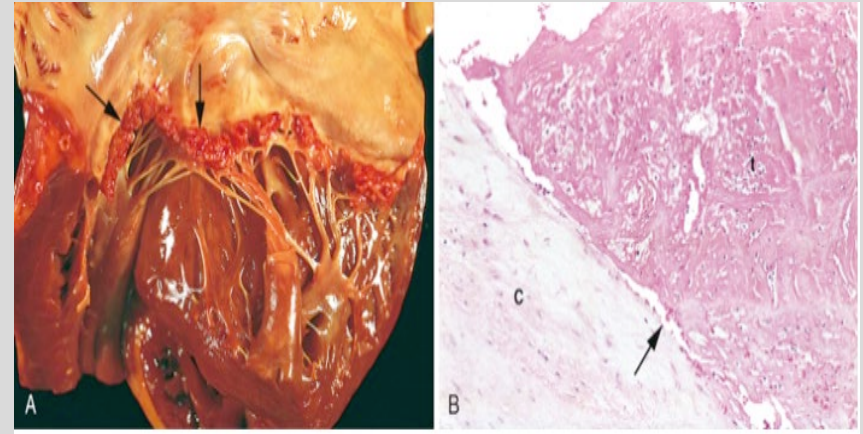
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Valvular damage is not a prerequisite for NBTE; this condition usually occurs on previously **normal valves**.

NBTE is often seen in those with **cancer-particularly mucinous adenocarcinomas, sepsis or associated with hypercoagulable states**

- Disseminated intravascular coagulation
- Hyperestrogenic states

Endocardial trauma, as from an indwelling catheter, is a well-recognized predisposing condition.



A. Nearly complete row of thrombotic vegetations along the line of closure of the mitral valve leaflets (arrows)

B. Histology of bland vegetation, with virtually no inflammation in the valve cusp (c) or the thrombotic deposit (t). The thrombotic material (**fibrin, platelets, blood components**) is only loosely attached to the cusp (arrow)

Libman-Sacks Endocarditis

Small (1-4 mm), sterile vegetations in patients who have systemic lupus erythematosus. Vegetations are due to immune complex deposition with activation of complement.

Vegetations can occur on any valve surface, chordae, or on the atrial and ventricular endocardium
Can lead to fibrosis, scarring and fusion of leaflets



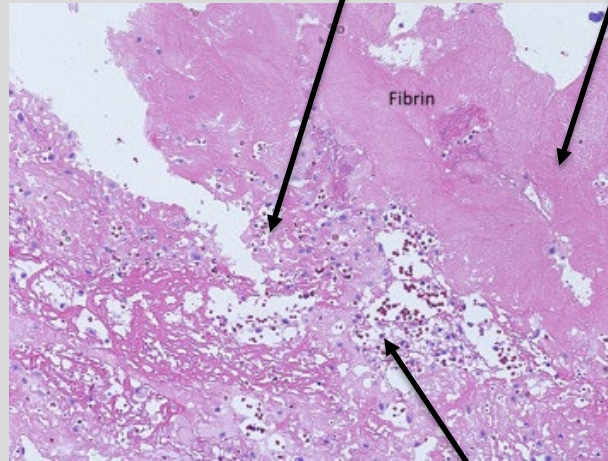
Vegetations

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Histology: intense **valvulitis** and **fibrinoid necrosis**: fine, granular, pink-tan, fibrinous eosinophilic material



[https://www.annalsthoracicsurgery.org/article/S0003-4975\(17\)30185-6/fulltext](https://www.annalsthoracicsurgery.org/article/S0003-4975(17)30185-6/fulltext)

Photomicrograph shows section of mitral valve leaflet vegetation with deposition of fibrin and small collections of **neutrophils and monocytes**, but no bacterial colonies

Summary

General features defining cardiac valvular disease

Aortic Stenosis

- Congenital bicuspid valve
- Acquired
- Equivalent of the age-related atherosclerosis
- "wear and tear" mechanism

Mitral Valve Prolapse

- floppy leaflets, myxomatous degeneration

Mitral Annular Calcification

- most of the time no therapy, associated with coronary and atherosclerotic disease.

Acute Rheumatic Fever vs Rheumatic Heart Disease

- group A beta hemolytic streptococci
- Aschoff bodies, Anitschikow cells
- ARF JONES major criteria
- RHD-the only cause of acquired mitral stenosis "fish-mouth" pattern (please also see table on slide 4)

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Infective Endocarditis

- acute vs subacute
- etiology, morphology, clinico-pathological consequences.
- Streptococcus viridans vs Staphylococcus aureus
- Janeway lesions, Roth spots, Nail bed or Splinter hemorrhages, Osler nodules

Nonbacterial thrombotic endocarditis

- mucinous adenocarcinoma, hypercoagulable states, sterile vegetations/thrombi, infarcts

Libman-Sacks endocarditis

- SLE, immune complex depositions, valvulitis, fibrinoid necrosis.

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